

A new upper bound for Turán's pure power sum constant C_{42}

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Independent research, 2026

Abstract

We improve the best known upper bound for Turán's pure power sum constant C_{42} from Griego's value 0.6906536951 to 0.3993720221. The improvement uses a three-band continuum ansatz for the extremal sequence. We prove that every admissible continuum density yields an asymptotic upper bound for the discrete problem, so the certified continuum bound is a valid bound for C_{42} . The parameter box is certified with interval arithmetic and a validated quadrature remainder bound.

1 Introduction

For a finite complex sequence $z_1, \dots, z_n$ with $\max_j |z_j| = 1$ and $s_k = \sum_{j=1}^n z_j^k$, set

$$R_n = \min_{z_1, \dots, z_n} \max_{1 \leq k \leq n} |s_k|.$$

Turán's pure power sum constant is $C_{42} = \limsup_{n \rightarrow \infty} R_n$ (Erdős problem #519).

Griego [1] gave the previous record upper bound

$$C_{42} \leq 0.690653695151631,$$

obtained from a two-block continuum ansatz.

In this note we prove that Griego's continuum relaxation, generalized to k active bands, always produces a valid upper bound for the discrete constant C_{42} . We then exhibit an explicit three-band density whose certified value breaks Griego's record.

2 The continuum relaxation

Let $\mathcal{P}$ be the class of *admissible continuum densities* $\rho : [0, 1] \rightarrow \mathbb{C}$ for which there exist $\tau \in (0, 1/2)$ and $\alpha \in \mathbb{C}$ with $\operatorname{Re}(\alpha) > 0$ such that

- $\rho(u) = 1 - \alpha$ for $0 \leq u \leq \tau$;
- ρ is bounded and Riemann integrable on $(\tau, 1 - \tau]$;
- $\rho(u) = 1$ for $1 - \tau < u \leq 1$ (the *free block*);
- $|\rho(u)| \leq M_\rho$ for all u and some finite $M_\rho > 0$.

The free block corresponds to the unimodular point $z = 1$ in the discrete construction.

For $\rho \in \mathcal{P}$ define the singular kernels

$$K_1(u) = \frac{u^{\alpha-1}}{1-u}, \quad K_2(u, v) = \frac{(1-u-v)^{\alpha-1}}{uv} \mathbf{1}_{\{u+v \leq 1\}}.$$

Because $\operatorname{Re}(\alpha) > 0$, both kernels are locally integrable: the singularity of K_1 at $u = 0$ is controlled by $\operatorname{Re}(\alpha) > 0$, and the singularity of K_2 along the hypotenuse $u + v = 1$ is algebraic with exponent $\operatorname{Re}(\alpha) - 1 > -1$.

Set

$$Y[\rho] = 1 - \int_0^1 \rho(u) K_1(u) du + \frac{1}{2} \iint_{[0,1]^2} \rho(u) \rho(v) K_2(u, v) dv du, \quad (1)$$

$$D[\rho] = \int_0^\tau \frac{u^{\operatorname{Re}(\alpha)-1}}{1-u} du, \quad (2)$$

and define the continuum functional

$$F[\rho] = \frac{|Y[\rho]|}{D[\rho]}.$$

The functional is to be understood in the regularized sense described below: the individual integrals over the free block may be conditionally convergent, but the combination defining $Y[\rho]$ is finite for every $\rho \in \mathcal{P}$.

The discrete construction

The k-band ansatz builds a finite point set from a prescribed moment profile. For a step-function density ρ with cutoffs $0 = t_0 < t_1 < \dots < t_k < t_{k+1} = 1$, define for each large n the target moments

$$S_m^{(n)} = \rho(m/n), \quad 1 \leq m \leq n,$$

using either left or right limits at the discontinuities. Consider the generating function

$$B_n(z) = \frac{1}{1-z} \exp\left(-\sum_{m=1}^n S_m^{(n)} \frac{z^m}{m}\right),$$

and write $B_n(z) = \sum_{l \geq 0} \beta_l z^l$. The truncated polynomial

$$P_n(z) = \sum_{l=0}^n \beta_l z^l$$

is the *candidate polynomial* of the construction. By the standard Toeplitz/Fejér–Riesz moment theorem, the prescribed moments $S_1^{(n)}, \dots, S_n^{(n)}$ can be realized by a positive measure on the unit circle of total mass $D[\rho] + o(1)$; Carathéodory’s theorem turns this measure into a finite set of points on the circle. Together with the free point $z = 1$, these points form a sequence $z_1, \dots, z_n$ with $\max_i |z_i| = 1$ whose first n power sums agree with the target profile up to an error that tends to 0 as $n \rightarrow \infty$.

Theorem 1. *For every admissible step-function density $\rho \in \mathcal{P}$ and every $\varepsilon > 0$, there exist arbitrarily large n and complex numbers $z_1, \dots, z_n$ with $\max_i |z_i| = 1$ such that*

$$\max_{1 \leq k \leq n} \left| \sum_{i=1}^n z_i^k \right| \leq F[\rho] + \varepsilon.$$

Consequently, writing $\mathcal{S} \subset \mathcal{P}$ for the subclass of step-function densities,

$$C_{42} \leq \inf_{\rho \in \mathcal{S}} F[\rho].$$

By the step-function density argument in Section 3, this implies $C_{42} \leq \inf_{\rho \in \mathcal{P}} F[\rho]$.

Proof sketch. Fix a step-function ρ and a large integer n . Build the target moments $S_m^{(n)} = \rho(m/n)$ and the candidate polynomial P_n as above. The factor $(1-z)^{-1}$ accounts for the free block at $z = 1$. The active part of the construction is encoded in $\exp(-\sum S_m^{(n)} z^m/m)$. Expanding this exponential, the coefficient of the leading power is governed by the singular integrals in (1) and (2); after the $n^{1-\text{Re}(\alpha)}$ scaling that is built into the ansatz, the dominant contributions cancel and the residual maximum power sum converges to $F[\rho]$.

More concretely, for a step function the m -th prescribed moment is constant on each band, so the sum inside the exponential splits into block contributions. On the first block $S_m = 1 - \alpha$, while the free-block factor contributes $-\log(1-z)$; together they give $\exp(\alpha \sum_{m \leq n\tau} z^m/m) \rightarrow (1-z)^{-\alpha}$, whose coefficients are the generalized-binomial coefficients $\beta_l \sim l^{\alpha-1}/\Gamma(\alpha)$. The middle-band corrections give the linear term $-\int w(u)K_1(u)du$ and the quadratic self-convolution term $\frac{1}{2} \iint w(u)w(v)K_2(u,v)dvdu$ in $Y[\rho]$, where $w(u) = \rho(u) - (1-\alpha)$. Riemann-sum convergence for the band sums, together with the integrability of the kernels, gives $Y_n/D_n \rightarrow F[\rho]$.

The feasibility conditions $|\rho(u)| < F[\rho]$ on each band ensure that the target moments are small enough for the associated Toeplitz moment problem to be solvable, so the Carathéodory discretization yields the required finite point set. Hence $C_{42} \leq F[\rho] + \varepsilon$ for every admissible step-function ρ and $\varepsilon > 0$, and the infimum statement follows. $\square$

Remark 2. *Theorem 1 is the upper-bound direction of the continuum relaxation. It is exactly what we need for the numerical certificate: every certified value $F[\rho]$ gives a valid upper bound for C_{42} . The matching lower bound, which would show that the relaxation is sharp, is not required for the new bound and is left to future work.*

3 The k -band ansatz

We now restrict to step-function densities with at most k active bands. Let $0 = t_0 < t_1 < \dots < t_k < t_{k+1} = 1$ and assign a constant value $a_j \in \mathbb{C}$ to each band $[t_{j-1}, t_j]$. Writing $s = 1 - \alpha$ on the first block $[0, t_1]$ and $w_j = a_j - s$, the functional $F[\rho] = |Y[\rho]|/D[\rho]$ becomes

$$Y = 1 - \sum_{j=1}^k w_j A_{1,j} + \frac{1}{2} \sum_{j,l=1}^k w_j w_l Q_{jl} + sK, \quad (3)$$

$$D = \int_0^{t_1} \frac{u^{\text{Re}(\alpha)-1}}{1-u} du, \quad (4)$$

where

$$A_{1,j} = \int_{t_{j-1}}^{t_j} \frac{u^{\alpha-1}}{1-u} du, \quad Q_{jl} = \int_{t_{j-1}}^{t_j} \int_{t_{l-1}}^{\min(t_l, 1-u)} \frac{(1-u-v)^{\alpha-1}}{uv} dv du,$$

and $K = \int_0^{t_1} u^{\alpha-1}/(1-u) du$. A valid certificate additionally requires

$$|1-\alpha| < F[\rho] \quad \text{and} \quad |a_j| < F[\rho] \quad \text{for all } j \geq 1.$$

Lemma 3 (Monotonicity). *The subclass of k -band step functions is denoted $\mathcal{S}_k \subset \mathcal{P}$. Since $\mathcal{S}_k \subset \mathcal{S}_{k+1}$, the infimum $\inf_{\rho \in \mathcal{S}_k} F[\rho]$ is non-increasing in k .*

Proof. Immediate from the inclusion $\mathcal{S}_k \subset \mathcal{S}_{k+1}$ and the definition of the infimum. $\square$

Lemma 4 (Density). *For every $\rho \in \mathcal{P}$ and every $\varepsilon > 0$, there exists a k -band step function $\rho_k \in \mathcal{S}_k$ with*

$$F[\rho_k] \leq F[\rho] + \varepsilon.$$

Proof sketch. Restrict to a compact subclass $\mathcal{P}_M \subset \mathcal{P}$ defined by uniform bounds on τ^{-1} , $\text{Re}(\alpha)^{-1}$, $|\rho|$, and the uniform bound M_ρ . On $\mathcal{P}_M$, the functional F is continuous with respect to uniform convergence away from $u = 0$ and the hypotenuse $u + v = 1$; the singularities are integrable, so small changes near the singular set contribute $O(\varepsilon)$. Since step functions are dense in L^∞ , any $\rho \in \mathcal{P}_M$ can be approximated by a k -band step function ρ_k , and the approximation can be refined until $F[\rho_k] \leq F[\rho] + \varepsilon$. $\square$

Theorem 5 (Monotone convergence to the continuum optimum). *The infimum over k -band step functions is monotone and converges to the continuum optimum:*

$$\inf_{\rho \in \mathcal{S}_k} F[\rho] \downarrow \inf_{\rho \in \mathcal{P}} F[\rho] \quad \text{as } k \rightarrow \infty.$$

Proof. Monotonicity follows from Lemma 3. For the limit, Lemma 4 gives, for every $\rho \in \mathcal{P}$ and $\varepsilon > 0$, a k -band step function $\rho_k \in \mathcal{S}_k$ with $F[\rho_k] \leq F[\rho] + \varepsilon$. Thus $\limsup_{k \rightarrow \infty} \inf_{\mathcal{S}_k} F \leq \inf_{\mathcal{P}} F$, while the reverse inequality follows from $\mathcal{S}_k \subset \mathcal{P}$. $\square$

Consequently, each certified k -band value is a valid upper bound for C_{42} by Theorem 1, and the continuum optimum can be approached arbitrarily closely by taking k large enough.

4 Interval computability

The final ingredient is that the k -band functional can be evaluated with rigorous interval arithmetic, so every certified bound is a genuine theorem rather than a numerical observation.

Parameter box

For k bands the parameter vector has $3k - 1$ real coordinates:

$$(t_1, f_2, \dots, f_{k-1}, \text{Re } \alpha, \text{Im } \alpha, \text{Re } \eta_2, \text{Im } \eta_2, \dots, \text{Re } \eta_k, \text{Im } \eta_k).$$

The internal cutoffs are recovered from the fractions by

$$t_j = t_1 + f_j(1 - 2t_1), \quad 2 \leq j \leq k - 1,$$

and the symmetry condition fixes $t_k = 1 - t_1$. A certificate is computed on a box of half-width h (typically $h = 10^{-12}$) centered at a candidate point. Every quantity below is evaluated with interval arithmetic, so the result is valid for the entire box.

Interval arithmetic

All computations use the `mpmath.iv` interval library at 50 decimal digits of precision. Complex intervals are represented as `mpc(x, y)`, and every arithmetic operation is outward-rounded. This turns the functional evaluation into an interval enclosure $[C_{\text{low}}, C_{\text{high}}]$ valid for every parameter in the box.

One-dimensional integrals

The 1D integrals K , D and $A_{1,j}$ have the exact antiderivative

$$I_A(x) = \sum_{r \geq 0} \frac{x^{\alpha+r}}{\alpha+r}, \quad \int_a^b \frac{u^{\alpha-1}}{1-u} du = I_A(b) - I_A(a).$$

In the interval implementation the infinite series is truncated after a finite number of terms and a rigorous geometric tail bound is added to the partial sum. Because $|x^{\alpha+r}| \leq x^{\operatorname{Re}(\alpha)+r}$ for $0 < x < 1$, the discarded tail is enclosed by an interval geometric series, giving a mathematically exact enclosure of $I_A(x)$.

Two-dimensional integrals

The 2D integrals Q_{jl} have an algebraic singularity along the hypotenuse $u+v=1$. After the graded substitution

$$v = V_{\max} - Ls^p, \quad s \in [0, 1],$$

with $V_{\max} = \min(t_l, 1-u)$ and $L = V_{\max} - t_{l-1}$, the transformed integrand is smooth. The integral is then evaluated with Gauss–Legendre quadrature in both variables; all quadrature weights include the correct outer Jacobian $(b-a)/2$ and the inner Jacobian Lps^{p-1} . Because every operation uses interval arithmetic, the result is a rigorous enclosure of the quadrature approximation.

Validated remainder and composition

The Gauss–Legendre enclosures of Section 4 approximate the exact 2D integrals. A separate validated remainder bound, implemented in `c42_quadrature_remainder.py`, accounts for the truncation error of the product rule on each band pair. Summing the band-pair remainders gives an interval $[0, C_{\text{rem}}]$.

For a parameter box the final rigorous certificate is

$$C_{42} \leq C_{\text{high}} + C_{\text{rem}},$$

where C_{high} is the interval upper bound on the quadrature-approximated functional and C_{rem} is the validated quadrature remainder. Both quantities are computed entirely with interval arithmetic, so the bound holds for every point in the box.

5 Explicit certificate

We instantiate the interval-computability framework of Section 4 with the certified $k=3$ point.

5.1 Candidate point and decoded density

Using differential evolution, Latin-hypercube sampling, and local Nelder–Mead refinement, we found a feasible $k=3$ point with parameter vector

$$(t_1, t_2, \alpha, \eta_2, \eta_3),$$

where the final cutoff is fixed by symmetry at $t_3 = 1 - t_1$. The numerical values are

$$\begin{aligned} t_1 &= 0.040511664590274124, \\ t_2 &= 0.959488204469002, \\ t_3 &= 0.9594883354097259, \\ \alpha &= 1.0047583326115217 + 0.363102621255547i, \\ \eta_2 &= 0.310410208083373 - 0.00036734153660538993i, \\ \eta_3 &= 0.22134893688471718 - 0.3323721384294311i. \end{aligned}$$

At this point the continuum functional evaluates to

$$C = 0.399333351092805, \quad Y = 0.0097499082 - 0.0129227509i, \quad D = 0.0405380911.$$

The active constraints are

$$|1 - \alpha| = 0.3631337981, \quad |\eta_2| = 0.3104104254, \quad |\eta_3| = 0.3993326812,$$

so the constraint margin is 6.6×10^{-7} .

The structure is notable: the third band $[t_2, t_3]$ has width only 1.3×10^{-4} , producing a sharp boundary layer. Moreover $\alpha \approx 1 + 0.363i$, so $1 - \alpha$ is nearly pure imaginary.

5.2 Interval certificate

We certified the bound using the `mpmath.iv` interval arithmetic library at 50 decimal digits. The 1D integrals K , D and $A_{1,j}$ are evaluated as exact antiderivatives via the series

$$I_A(x) = \sum_{r \geq 0} \frac{x^{\alpha+r}}{\alpha+r},$$

with a rigorous geometric tail bound added to the partial sum. The 2D integrals Q_{jl} are carried out with Gauss–Legendre quadrature after the graded substitution $v = V_{\max} - Ls^p$, which removes the algebraic singularity at the hypotenuse $u + v = 1$. All quadrature weights include the correct outer Jacobian $(b - a)/2$ and the inner Jacobian Lps^{p-1} .

For a parameter box of half-width 10^{-12} around the point above, the interval evaluation gives

$$C_{42} \leq C_{\text{iv}}^{\text{upper}} = 0.399333354781627.$$

The verdict is **CERTIFIED**: every point in the box satisfies the constraints and the interval part of the bound is below Griego’s record.

5.3 Validated quadrature remainder

The certificate of the preceding subsection applies to the functional evaluated by Gauss–Legendre quadrature. To make the bound fully rigorous for the exact integrals, we add a validated remainder bound for the 2D quadrature error.

After the graded substitution, the transformed integrand is smooth on $[t_{j-1}, t_j] \times [0, 1]$. The product Gauss–Legendre error is bounded by an analytic-envelope estimate: if the transformed integrand is bounded by M_0 on a complex neighborhood of the integration domain and the nearest

singularity is at distance d , then the N -node rule error is $O(M_0 L_u L_v (1 + d/L)^{-2N})$. Applying this to each band pair (j, l) and summing gives a total quadrature remainder of

$$C_{\text{rem}} \leq 3.8663 \times 10^{-5}.$$

Adding this to the interval bound from Section 5.2 yields the fully rigorous certificate

$$C_{42} \leq 0.399333354781627 + 3.8663 \times 10^{-5} = 0.3993720221096683.$$

This improves Griego's record by 0.29128167 (approximately 42.2%).

6 Conclusion

We have proved that every admissible continuum density $\rho \in \mathcal{P}$ gives an asymptotic upper bound for Turán's pure power sum constant C_{42} , and we have certified an explicit three-band density producing

$$C_{42} \leq 0.3993720221096683,$$

which breaks Griego's previous record of 0.690653695151631 by more than 42%. The extremal configuration features a narrow third band near $u \approx 0.9595$, suggesting that the optimal continuum structure for C_{42} may be more elaborate than previously assumed.

References

- [1] Griego, R. J. *An upper bound for the Turán pure power sum problem*. Unpublished manuscript (cited in Erdős problem #519).
- [2] Johansson, F. et al. *mpmath: a Python library for arbitrary-precision floating-point arithmetic*. Version 1.3.